RITIKA CHANDRAKANT BOBHATE

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Software Engineering | AI/ML • Data Analytics

EDUCATION

VIDYALANKAR INSTITUTE OF TECHNOLOGY- Bachelor of Technology — Mumbai, MH
Electronics and Computer Science; Majors : Data Analytics — Expected by 2026
CGPA: 9.07/ 10.0;

WORK EXPERIENCE

RELIANCE INDUSTRIES PVT LTD - Software Development Intern — Navi Mumbai, MH | Dec 2025 – Jan 2026

- Implemented a full-stack RCA platform using React and MongoDB Atlas supporting analysis of 100+ simulated failure cases.
- Collaborated with mentors to design, implement, and validate the system in an industrial diagnostics workflow.

SUVIDHA FOUNDATION – Data Science Research Intern — **Mumbai, MH** | Apr 2025 – Jun 2025

Cross-Lingual Commonsense Reasoning Using XLM-R — IEEE REACS 2025
DOI: https://doi.org/10.1109/REACS67479.2025.11413321

- Performed multilingual text preprocessing, feature extraction, and model evaluation in Python for cross-lingual NLP analysis.
- Collaborated with research mentors to design experiments, validate results, and prepare the research manuscript using LaTeX.

TRINETRINI QUANTUM PVT LTD – Web Development Intern — **Mumbai, MH** | Jun 2024 – Jul 2024

- Designed web UI features using Zoho CRM, reducing user navigation confusion by 20% and improving UX.

PROJECTS

Confidence-Driven Hybrid ML-LLM Framework for Automated System Failure Root Cause Analysis
Tech: Python, Groq API (LLM), React, FastAPI, MongoDB Atlas, Docker, JWT | — Jan 2026
https://github.com/RitikaBobhate/Confidence-Driven-Hybrid-ML-LLM_Framework-for-Automated-System-Failure-Root-Cause-Analyzer

- Developed a hybrid ML + LLM system for industrial root cause analysis using the 5-Why methodology, achieving 80%+ prediction confidence.
- Implemented a full-stack platform with React, FastAPI, and MongoDB Atlas, decreasing the time to identify root causes by approximately 25% through interactive visualizations.

URBAN THERMAL MAPPING USING SATELLITE IMAGERY - (Ongoing) — Jan 2026
Tech: QGIS, YOLOv5, PolsarPro, LabelImg, Power BI |
https://github.com/RitikaBobhate/Urban-Heat-Island-Detection-using-YOLOv8

- Processed Landsat satellite imagery using QGIS and created labeled datasets with LabelImg for urban heat zone detection.
- Integrated a YOLOv5-based model to identify high-temperature regions and visualized heat distribution through interactive Power BI dashboards.

LuxeEstimate – SMART HOUSE PRICE — Jan 2026
Tech: Python, Random Forest, Streamlit, RestAPI, Data Cleaning. |
https://github.com/RitikaBobhate/-LuxeEstimate-Smart-House-Price-Evaluation

- Created a machine learning model to predict house prices using Random Forest, achieving R^2 scores of 0.90 (train) and 0.80 (test).
- Performed feature engineering and constructed a responsive Streamlit web application for real-time property price estimation.

ACTIVITIES

IEEE GRSS COUNCIL - Creatives Co-Head | Feb 2024 – Feb 2025 — VIT, MH
● Orchestrated comprehensive marketing campaigns for GRSS technical workshops, using targeted email marketing, social media, and digital advertising, resulting in a 40% surge in student participation rates.

TECHNICAL COMMITTEE — STUDENT COUNCIL, VIT — VIT, MH
● Deployed the official Verve college website (received college recognition) and coordinated a technical committee event.

ADDITIONAL

Technical Skills: Python, SQL, Java, OpenCV, YOLOv5, Scikit-learn, Pandas, NumPy, Matplotlib, Power BI, React, FastAPI, Flask, MongoDB, MySQL, QGIS, Docker, Git, LaTeX, UiPath (RPA), Excel.

Certifications: Supervised & Unsupervised Machine Learning — Andrew Ng (DeepLearning.AI/Stanford Online) | Google Data Analytics — Coursera | Python for Data Science — IBM | Data Analysis & Visualization using Python — CDAC ACTS | Big Data Analytics — Great Learning